

Light Source Selection in Primary-Sample-Space Neural Photon Sampling

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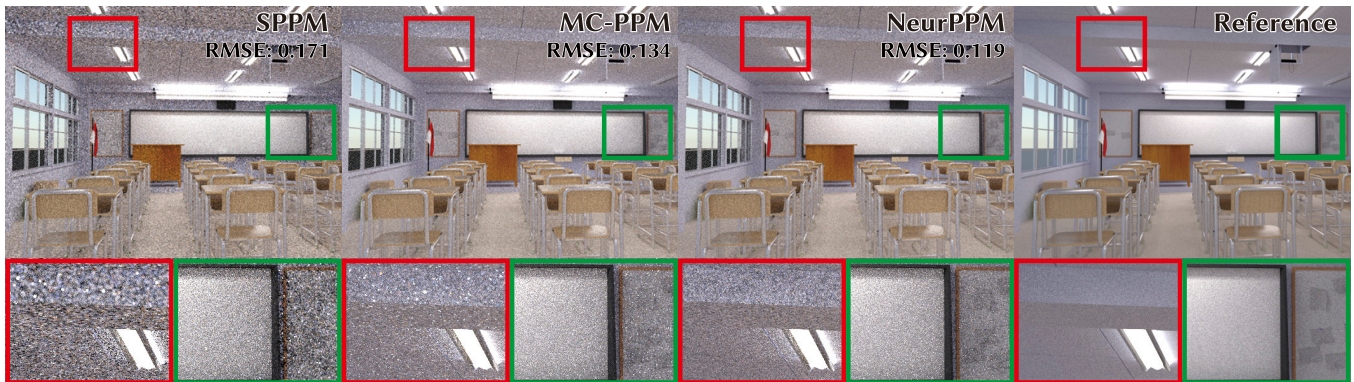


Figure 1: Equal sample comparison (with 200 million photons) of stochastic progressive photon mapping (SPPM) [Hachisuka and Jensen 2009], photon sampling using Markov chain Monte Carlo (MC-PPM) [Hachisuka and Jensen 2011], and our method with the reference image obtained by bidirectional path tracing (BDPT) with 1024 samples per pixel (spp). As shown by the enlarged parts, our method has successfully suppressed fireflies to achieve lower variance than the previous methods.

ABSTRACT

This paper proposes a light source selection for photon mapping combined with recent deep-learning-based importance sampling. Although applying such neural importance sampling (NIS) to photon mapping is not difficult, a straightforward approach can sample inappropriate photons for each light source because NIS relies on the approximation of a smooth continuous probability density function on the primary sample space, whereas the light source selection follows a discrete probability distribution. To alleviate this problem, we introduce a normalizing flow conditioned by a feature vector representing the index for each light source. When the neural network for NIS is trained to sample visible photons, we achieved lower variance with the same sample budgets, compared to a previous photon sampling using Markov chain Monte Carlo.

KEYWORDS

photon mapping, importance sampling, neural networks

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1 INTRODUCTION

Recently, two concurrent studies [Müller et al. 2019; Zheng and Zwicker 2019] proposed neural importance sampling (NIS), i.e., importance sampling using a neural network, for Monte Carlo path tracing, having improved the efficiency of importance sampling with the same sample budgets. NIS trains a neural network called the normalizing flow (NF) to transform a random sample following a probability distribution function (PDF) without an analytic form to the one following another closed-form PDF. The invertibility of the NF helps random sampling from a non-analytic PDF by inversely transforming the sample from the closed-form PDF.

This paper introduces NeurPPM, an application of NIS to photon mapping, which is based on stochastic progressive photon mapping (SPPM) [Hachisuka and Jensen 2009] but samples photons using NIS. NeurPPM trains the NF to sample visible photons as in a previous study using Markov chain Monte Carlo (MCMC) [Hachisuka and Jensen 2011]. Although this extension of SPPM using NIS is not difficult, such a simple extension can sample inappropriate photons for each light source. To address this problem, we introduce a normalizing flow conditioned by a feature vector representing the light index. Our method improves the efficiency of visible photon sampling significantly, achieving lower variance with the same sample budgets than a previous approach using MCMC.

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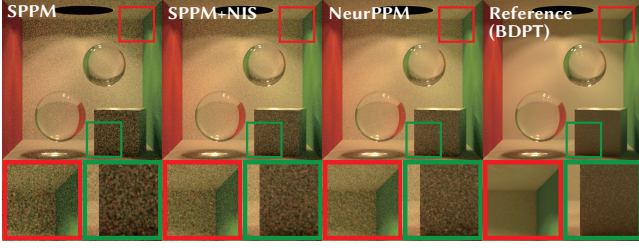


Figure 2: Compared to the results of a simple combination of SPPM and NIS, the amount of noise is significantly reduced by our NeurPPM using feature vectors for a light index.

2 NEURAL PHOTON SAMPLING

For NIS in Monte Carlo path tracing, we collect samples to train the NF by standard path sampling using uniform random vectors $\mathbf{u}$ from the primary sample space (PSS) $\bigcup_{d=1}^{\infty} [0, \infty)^d$. Then, a set of random vectors $\mathbf{v} \sim P(\mathbf{v}) \propto I(x(\mathbf{v}))$ are resampled from the collected uniform samples, where $I(x(\mathbf{v}))$ is a contribution of the path $x(\mathbf{v})$ to the rendering image. Finally, the NF Φ is trained such that the NF transforms resampled random vectors to distribute uniformly over the PSS. The invertibility of the NF allows us to sample $\mathbf{v} \sim P(\mathbf{v})$ by $\mathbf{v} = \Phi^{-1}(\mathbf{u})$ where $\mathbf{u}$ is a uniform random vector from the PSS. In practice, the NF is trained using only the first several dimensions of the random vector $\mathbf{u}$ because the neural network cannot learn random vectors with infinite dimensions. We write the first D dimensions of $\mathbf{u}$ as $\mathbf{u}_0 \dots \mathbf{u}_D \in \mathbb{R}^D$.

NeurPPM uses NIS for sampling photon paths while sampling camera paths by traditional importance sampling. However, as shown in Fig. 2, a simple application of NIS to photon sampling does not obtain visible photons significantly more than SPPM, even in the case where the scene consists of only nine lights. The same problem is observed in a simpler experiment to approximate Gaussian distribution. In Fig. 3, the target distribution consists of multiple 1D Gaussians along the vertical axis, each of which is defined over a section of the second dimension. When the NF approximates this 2D distribution, the Gaussian in a section is leaked to adjacent sections, as shown in the left column. We can avoid this problem by converting the random variable along the horizontal axis to an index of the section and encoding the index to a feature vector conditioning the NF. Following the result of this experiment, we use the feature vector to represent the light index rather than selecting a light source using the first dimension of the random vector $\mathbf{u}_0 \dots \mathbf{u}_D$.

A light index $\ell \in \mathbb{N}$ is calculated by $\ell(u_0) = \lfloor u_0 K \rfloor$ using the number of light sources K . The feature vector for ℓ is obtained by an index encoder $E: \mathbb{N} \rightarrow \mathbb{R}^F$ that involves a lookup table mapping the index to a predefined random vector, followed by a fully-connected layer. In our experiment, we transform the index to a feature vector of $F = 8$ dimensions. With this feature vector, the NF transforms the random vector $\mathbf{u}_1 \dots \mathbf{u}_D$ without the first dimension. As with the ordinary NIS, this conditional NF is trained to transform a pair of resampled random vector $\mathbf{v}_1 \dots \mathbf{v}_D$ and light index ℓ to a uniform random vector in $\mathbb{R}^{D-1}$. Then, we can importance-sample visible photons using a random vector $\mathbf{v}_1 \dots \mathbf{v}_D = \Phi^{-1}(\mathbf{u}_1 \dots \mathbf{u}_D, E(\ell(u_0)))$ with a uniform random vector $\mathbf{u}_0 \dots \mathbf{u}_D$.

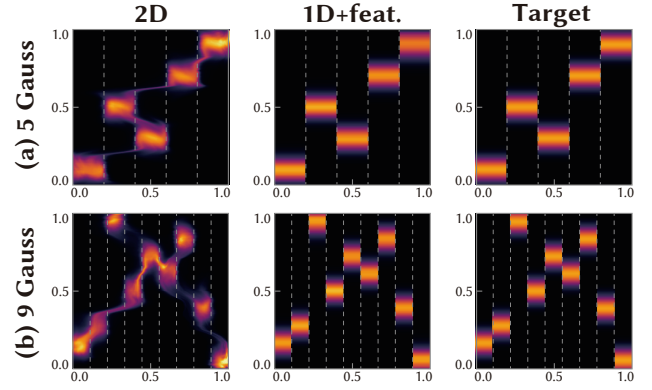


Figure 3: A target distribution in the right is not well represented by an NF when it is handled as a 2D distribution, whereas it is done significantly well when a random variable for the horizontal axis is converted to a feature vector.

3 RESULTS AND DISCUSSION

Our method is implemented on top of PBRT v3 [Pharr et al. 2016]. We construct a training dataset by shooting 20 million photons and collecting random vectors to obtain visible photons. We used Real NVP [Dinh et al. 2016] as the NF, trained over 5120 steps using mini-batches of 2000 samples, and is optimized by Adam optimizer with learning rate $\gamma = 0.0001$.

Figure 1 compares the high-resolution result (1920×1080 pixels) of NeurPPM with those of SPPM [Hachisuka and Jensen 2009] and MC-PPM [Hachisuka and Jensen 2011]. The results for these methods are rendered using 200 million photons, and the reference image is rendered by BDPT using 1024 spp. Comparing the two enlarged regions of each image, NeurPPM has successfully suppressed fireflies and achieved lower variance than the previous approaches. We included results for other scenes in the supplementary materials.

Unfortunately, our method for photon mapping involves the same limitation as NIS for path tracing. Specifically, the variance decreases when the same “sample” budgets are used for comparison because path/photon sampling by a neural network takes more time than ordinary random number generators. For instance, it spent NeurPPM about 8 minutes to render the image in Fig. 1 while SPPM and MC-PPM spent 1.5 and 2.5 minutes, respectively.

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